

GOGUL KUMAR MATHI

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AI Data Scientist II driving transformation through intelligent automation and predictive analytics. Delivered measurable impact reducing C-suite operational burden and enabling real-time executive decision-making across finance, healthcare, and life sciences. Expert at translating complex business challenges into data-driven AI insights, automated critical earnings workflows, and deployed forecasting models improving liquidity and investment planning.

WORK EXPERIENCE

Data Scientist II || Expedia Group

Nov, 2024 - Present

- Automated Expedia earnings preparation for CFO/CEO by deploying agentic AI system that orchestrates script generation, competitive analysis, sentiment tracking, and source citation across **2K+** quarterly reports—reduced analyst workload from **5 days to 2 days (60% time savings)** and improved executive confidence with grounded, market-aware earnings narratives.
- Developed ODIN, a **LangGraph**-based agentic finance intelligence chatbot for Expedia Corp Finance Group, orchestrating agentic workflows to answer business questions across revenue, spend, and performance metrics enabled finance analysts to generate insights **70%** faster with **90%** SQL accuracy, **<3s** latency, and **4.5/5** user adoption
- Built an **LLM-powered presentation automation engine** with narrative generation, template rendering, and multi-turn conversational memory, automating **50+ executive decks monthly**. Developed a validation layer for factual accuracy and brand compliance, cutting creation time from **4 days → 10 minutes**.
- Built predictive cash flow models for Expedia Treasury Department to forecast **weekly/monthly** direct cash flows critical for liquidity management and investment planning, deploying ensemble time-series (**SARIMA, Holt Winters**) and ML techniques (**XGBoost**, Random Forest) across booking patterns, transaction volumes, and payment timing data from 4 global brands—improved accuracy by **20%**
- Analyzed marketing ad revenue data using statistical modeling on **CPC, CTR**, and **impressions** to optimize ad strategy, driving 2% lift in performance and informing marketing investment decisions

Data Scientist || New York State Department of Health

Mar, 2024 - Nov, 2024

- Interpreted, analyzed public health vital records data to identify trends, insights from key metrics like COD, Gestation time facilitating data-driven decisions using **Snowflake** and **Tableau** dashboards
- Managed data inconsistencies, ensuring data accuracy and reliability for improved reporting to government officials
- Conveyed actionable insights through, visualizations, enhancing understanding and reporting of healthcare facilities

Data Scientist Intern|| Cerevel Therapeutics

Jun, 2022 - Dec, 2022

- Slashed drug metabolism(DMPK) analysis time by **75%**, by analyzing, interpreting key clinical metrics of different species via **Tableau** dashboards
- Developed a real-time data pipeline integrating **Snowflake** with **AWS Kinesis Firehose**, and partnered with a cross-functional team to perform real-time behavior analysis of trends and anomalies in clinical DMPK data
- Minimized clinical textual data review time by **50%** through NLP (Spacy, Stanford-NLP, NLTK, and **SciSpacy**), identifying key clinical terms to predict overall drug sentiment, achieving 0.85 f1-score

Decision Scientist || MuSigma Business Solutions

Jun, 2019 - Aug, 2021

- Boosted underwriting efficiency by 10% by visualizing Financial Hardship metrics and aiding stakeholders in strategic decisions via interactive [Tableau](#) dashboards
- Built a large-scale **AWS Redshift data** warehouse (ELT) analytics layer from on-premises **IBM DB2 OLTP insurance claims** data using **SQL**
- Managed big data CI/CD pipeline **orchestration** with **Jenkins**, maintained version control using **Bitbucket**. Utilized **Jira** for project tracking and documented all phases in **Confluence**, ensuring seamless collaboration and process clarity

- Performed A/B testing to identify optimal target audience and offer mix for a large-scale acquisition campaign, reducing cost-per-acquisition by 15% and helping suppliers understand and acquire their most loyal customers

SKILLS

- **Programming Languages:** Python (NumPy, Pandas, SciPy, TensorFlow, Keras, Matplotlib, Seaborn, Plotly)
- **BI Tools:** Tableau (LODs), Power BI, Looker, Flourish, Datawrapper, Microsoft Excel
- **Database Languages:** MySQL, NoSQL, Hive, Hadoop, Kafka, Spark SQL, Microsoft SQL Server
- **Data Engineering Skills:** Data Ingestion, Data Warehousing, Data Marts, ETL/ELT,
- **Warehousing Tools:** Snowflake, Azure Synapse Analytics, AWS Redshift, Google BigQuery
- **Data Processing Tools:** AWS Lambda, Azure Data Factory, Google Dataflow
- **LLM Frameworks :** LangChain, Prompt Engineering, LLM-based Automation, Text Summarization, Sentiment Analysis, RAG Pipelines, Streamlit for AI apps
- **Prompt Engineering:** Few-shot learning, chain-of-thought prompting, system prompt optimization, context engineering
- **Fine-tuning & Customization:** LoRA, QLoRA, RLHF, instruction tuning, parameter-efficient fine-tuning
- **Retrieval Augmented Generation (RAG):** Multi-stage RAG, hybrid search, semantic chunking, reranking strategies

EDUCATION

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|---|----------------------------------|
| ■ Northeastern University, Boston, Massachusetts | Graduation Date - 12/2023 |
| Master of Science, Data Science | GPA 3.91/4.0 |